

LPG Gas Agency Management System - Project Blueprint

Comprehensive Software Development Plan

Problem Statement

Develop an end-to-end LPG agency platform consisting of Customer Mobile App, Driver Mobile App, and Agency Web Dashboard. The solution manages booking, delivery, cylinder inventory, customer balances, accounting, sales, and reporting.

Scope

Customer booking, order tracking, payments, complaints, delivery management, cylinder inventory tracking, accounting, reporting, analytics, and administration.

Technology Stack

.NET 8 ASP.NET Core APIs, SQL Server, Flutter (Android/iOS/Web), Azure Cloud, Azure DevOps CI/CD.

Core Modules

Customer Management, Order Management, Delivery Management, Inventory Management, Cylinder Ledger, Accounting, Sales, Reporting, Notifications.

Cylinder Inventory Logic

Track filled and empty cylinders at warehouse, vehicle, and customer levels. Maintain customer-wise ledger showing filled delivered, empty collected, current balances, and historical transactions.

Technical Architecture

Flutter Apps -> API Gateway -> ASP.NET Core Services -> SQL Server. Supporting services include Authentication, Notification Service, Reporting Engine, and Payment Gateway.

Phase 1

Business Analysis, Requirement Gathering, Process Mapping, Wireframes, Use Cases, Data Models.

Phase 2 (Advanced Features)

WhatsApp booking integration, AI demand forecasting, route optimization, QR/Barcode cylinder tracking, customer consumption analytics, predictive inventory planning, distributor integrations, geo-fencing, eKYC, chatbot support, BI dashboards.

Deployment

Azure App Services, Azure SQL Database, Blob Storage, Monitoring, Backup Strategy, Disaster Recovery, Load Balancer, CI/CD pipelines.

Team Structure

Project Manager, Business Analyst, Solution Architect, UI/UX Designer, 2 .NET Developers, 2 Flutter Developers, 2 QA Engineers, DevOps Engineer.

Estimated Timeline

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This is essentially an LPG Distribution Management System with three user groups: Customers, Agency Staff, and Delivery Drivers.

Problem Statement

Most LPG agencies still manage bookings, deliveries, customer cylinder balances, and inventory using disconnected systems or manual processes. This creates:

- Poor visibility of filled and empty cylinders
- Delivery tracking issues
- Inventory mismatches
- Delayed accounting and billing
- Customer service challenges
- Lack of real-time reporting

The proposed solution will digitize the complete LPG cylinder lifecycle from booking to delivery and inventory reconciliation.

Scope of Application

1. Customer Mobile App (Android & iOS)

Features:

- Customer registration/login
- OTP authentication
- View profile
- View cylinder balance
- Book refill cylinder
- Track delivery status
- Delivery history
- Invoice download
- Online payment integration
- Complaint management
- Notifications and reminders

2. Agency Web Dashboard

Customer Management

- Customer master
- KYC management
- Customer cylinder inventory
- Consumer number mapping

Order Management

- New bookings
- Pending orders
- Assigned orders
- Delivered orders
- Cancelled orders

Inventory Management

- Filled cylinder stock
- Empty cylinder stock
- Warehouse inventory
- Vehicle inventory
- Customer inventory

Delivery Management

- Route planning
- Driver assignment
- Vehicle allocation
- Delivery monitoring

Accounting Module

- Sales invoices
- Payment collection
- Outstanding balances

- Cash collection
- GST reports
- Ledger reports

Reports

- Daily sales
- Cylinder movement
- Inventory reconciliation
- Driver performance
- Customer consumption analysis

Dashboard KPIs

- Orders today
 - Deliveries completed
 - Revenue
 - Filled stock available
 - Empty cylinders collected
 - Outstanding payments
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3. Delivery Driver Mobile App

Delivery Operations

- Login
- View assigned deliveries
- Route navigation
- Customer location

Inventory Tracking

At delivery:

Driver loads:

- 50 Filled Cylinders
- 10 Empty Cylinders

After delivery:

Example:

- Delivered 15 Filled
- Collected 14 Empty

Vehicle stock automatically updates:

- Filled = 35
- Empty = 24

Delivery Confirmation

- OTP verification
- Customer signature
- Photo proof
- GPS capture

Payment Collection

- Cash
- UPI
- Card

Core Business Logic

Customer Cylinder Ledger

Maintain per customer:

Transaction	Filled	Empty
Initial	2	0
Refill Delivery	+1	-1
Empty Return	0	+1

Current balance:

- Filled with Customer
- Empty with Customer

Example:

Customer Inventory:

- Filled Cylinders = 2
- Empty Cylinders = 1

System should always know exact customer holdings.

This is the most important module.

Recommended Technology Stack

Frontend

Mobile Apps

Flutter

Single codebase:

- Android
- iOS

Web Dashboard

- Flutter Web
- OR Angular

Recommended:

Flutter Web for common code sharing.

Backend

API Layer

ASP.NET Core

Features:

- REST APIs
 - JWT Authentication
 - Role-based security
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Database

Microsoft SQL Server

Tables:

- Customers
- Orders
- Deliveries
- Vehicles
- Drivers
- Inventory
- Cylinder Transactions
- Payments
- Invoices

Cloud Infrastructure

Option 1 (Recommended)

Microsoft Azure

Components:

- App Service
- Azure SQL
- Blob Storage
- Notification Hub

Option 2

Amazon Web Services

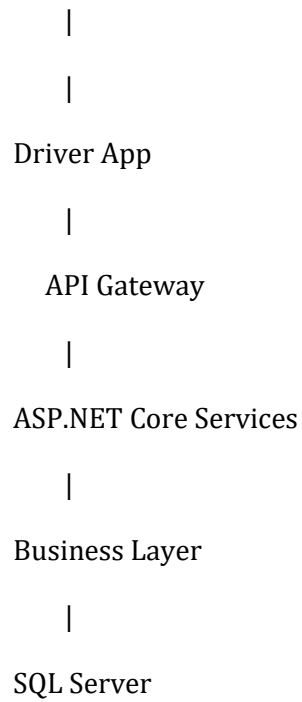
High-Level Architecture

Customer App

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Agency Dashboard



Development Phases

Phase 1 - Business Analysis

Duration: 2 Weeks

Deliverables:

- Requirement gathering
 - Process mapping
 - Wireframes
 - Use cases
 - Data flow diagrams
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Phase 2 - System Design

Duration: 2 Weeks

Deliverables:

- Architecture document
- Database design
- API specifications

- Security design
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Phase 3 - UI/UX Design

Duration: 3 Weeks

Deliverables:

- Customer App designs
- Driver App designs
- Dashboard designs

Tools:

- Figma
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Phase 4 - Backend Development

Duration: 8–10 Weeks

Modules:

1. Authentication
 2. Customer Management
 3. Order Management
 4. Inventory Management
 5. Delivery Management
 6. Accounting
 7. Reporting
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Phase 5 - Mobile Development

Duration: 8 Weeks

Customer App

- Booking
- Tracking
- Payments

Driver App

- Delivery
 - Inventory updates
 - GPS tracking
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Phase 6 - Dashboard Development

Duration: 6 Weeks

Modules:

- CRM
 - Inventory
 - Sales
 - Accounting
 - Reporting
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Phase 7 - Testing

Duration: 4 Weeks

Testing Types:

- Unit Testing
 - Integration Testing
 - UAT
 - Security Testing
 - Load Testing
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Phase 8 - Deployment

Production Environment

- Load Balancer
- Application Server
- Database Server

- Backup Server

CI/CD

Use:

- Azure DevOps
- GitHub Actions

Suggested Team Structure

Role	Count
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Business Analyst	1
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Solution Architect	1
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UI/UX Designer	1
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.NET Developers	2
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Flutter Developers	2
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QA Engineers	2
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DevOps Engineer	1
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Project Manager	1
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Total Team:	10 People
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Estimated Timeline

Phase	Duration
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Analysis	2 Weeks
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Design	2 Weeks
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UI/UX	3 Weeks
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Development	12 Weeks
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Testing	4 Weeks
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Deployment	1 Week
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A future Phase 2 can include WhatsApp booking, AI-based demand forecasting, route optimization, barcode/QR cylinder tracking, and integration with government LPG distributors such as Indian Oil Corporation, Bharat Petroleum, and Hindustan Petroleum.